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PROJECT AND TEAM INFORMATION

Project Title

Smart Traffic Management System Using OS-Based Real-Time Process Scheduling

Student/Team Information

Team Name: Team #	StructMinds OSDBMS-V-2026-T004
Team member 1 (Team Lead)	Gaur, Ayush — 24021533 ayushgaur0119@gmail.com 
Team member 2	Manwal, Vedansh :24021697, vedanshmanwal@gmail.com 
Team member 3	Pujari, Anmol: 240212120 anmol.a1b2@gmail.com 

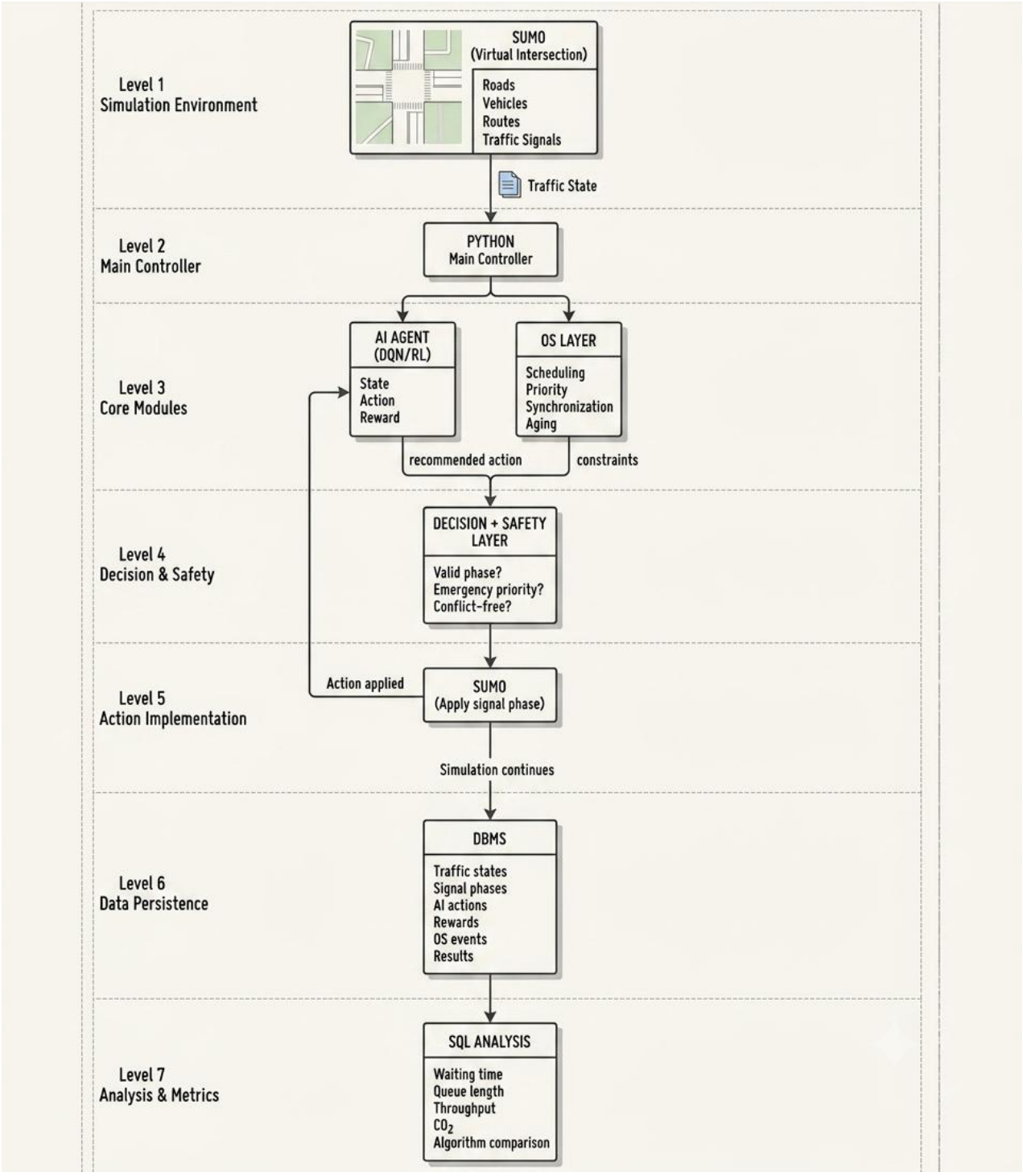
PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

Too often, we're stuck in traffic and even more so at the intersections of our communities, where cars enter into other flows of traffic at a seemingly constant rate. In fixed time traffic signals, the light is set to a specific amount of time for specific turn, so even when there is no cars there, the light can stay on the light for that turn, which increases the line in other directions. Our project is about how a traffik light will respond to dynamic traffik.
Traffic flow simulation at the four-way traffic intersection with traffic flow simulation is made by SUMO (Simulation of Urban Mobility) for Smart Traffic Management System demonstration. Interface of simulation and Python script is created by TraCI. The controller algorithm, embedded in the demo, relies on future information like waiting car in front of the signal or the number of queuing cars for phase selection.

Concepts from Operating Systems (priority, fair scheduling, safe scheduling, etc.) are being experimented with Python control layer. The database component is expected to store current traffic information, traffic signal change, traffic decision and performance measurements.
The single-intersection network has been created and viewed, the SUMO GUI is operational, and a fixed-time simulation is completed at this point. The adaptive controller, reinforcement-learning functionality, database logging, and complete comparison and evaluation are still the essential pieces to complete and verify.

Updated Project Approach and Architecture



Tasks Completed

Task Completed	Team Member
Problem identification and project planning	Vedansh Manwal
Python environment and SUMO setup	Ayush Gaur and Vedansh Manwal
Traffic network creation	Vedansh Manwal
Network verification and signal-phase inspection	Vedansh Manwal
Python–SUMO communication testing	Ayush Gaur
Fixed-time baseline simulation	Anmol Pujari
Preliminary performance metric collection	Anmol Pujari

Challenges/Roadblocks

The initial network-generation method needed to be modified to produce the desired 4-way network.
More detailed inspection was needed to understand the traffic-light phases and regulated motion of SUMO.
The demands for traffic in the simulation required simplification to make testing easier.
Baseline script required insertion of meaningful performance metrics.
It was reported as zero arrived vehicles for the baseline, so the route completion is requiring investigation to see if the results can be considered fully validated.
The combination of the OS scheduling, the DQN agent and database will still be a key thing of this project.

Tasks Pending).

Task Pending	Team Member (to complete the task)
Investigate route completion and validate baseline results	Vedansh Manwal
Implement OS-inspired scheduling and fairness rules	Vedansh Manwal
Develop and test the DQN controller	Ayush Gaur
Design and integrate the MySQL database	Anmol Pujari
Integrate all modules and test signal safety	Ayush Gaur and Vedansh Manwal
Compare different traffic-control strategies	Anmol Pujari
Final documentation and presentation	Vedansh Manwal

Project Outcome/Deliverables

The current implementation can:

- Form four ways crossroad network.

- Driving simulation in SUMO.

- Connect to SUMO from Python with TraCI.

- The inspection of traffic light phases and controlled links.

- Run 300 second fixed time baseline.

- Collect first stopped-vehicle observations.

The final deliverables will consist of the integrated traffic-control system, OS-inspired scheduling, DQN-based control, MySQL data storage, performance comparisons, and project documentation.

Progress Overview

Between 80-90% of the scope of the project is completed, which is comparable to the scope defined in this report (Phase-II). Simulation in traffic and the initial baseline are functioning; integration and validation of other modules are being done. Baseline was 1,457 stopped vehicle-seconds with a maximum of 14 vehicles stopped at one time. Zero-arrival outcome is not definitive and should be further examined. Final performance assessment will occur after the baseline and modules are validated.

Codebase Information

GitHub Repository: https://github.com/AyushGaur07/Smart-Traffic-Management-System-Using-OS-Based-Real-Time-Process-Scheduling
Branch: Main.
Team Contributions:

- **Ayush Gaur:** Project coordination and OS-based scheduling design.
- **Vedansh Manwal:** SUMO setup and traffic network creation.
- **Anmol Pujari:** Signal inspection and baseline testing.

The repository maintains the project code and development history. Individual commits can be checked in the GitHub commit history.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Python and SUMO setup	Pass	Environment successfully configured
Traffic network generation	Pass	Four-way intersection established on all sides
Junction verification	Pass	One controlled traffic-light junction located
GUI simulation	Pass	Observed vehicle motion
Signal-phase inspection	Pass	There are four kinds of signals phases
Fixed-time baseline execution	Pass	Simulation time of 300 seconds reached
Route-completion validation	In progress	Zero arrivals reported; investigation required
OS scheduling integration	Pending	Scheduling and fairness rules test required
DQN controller testing	Pending	AI controller needs to be implemented and tested.
MySQL integration	Pending	Database connection still persists
Full-system testing	In progress	Final integration and validation needed

Deliverables Progress

Deliverable	Status
Traffic Network	Completed
Python-SUMO Connection	Completed
Fixed-Time Baseline	Completed
OS Scheduling	In Progress
DQN Controller	Pending
MySQL Database	Pending
Performance Comparison	In Progress
Documentation	In Progress